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1 Data Analysis Expressions (DAX)

Data Analysis Expression (DAX) is the formula language used in Microsoft Power BI, Power Pivot, and SQL Server Analysis Services (SSAS) Tabular Models. DAX enables analysts and developers to create custom calculations, business metrics, KPIs, and advanced analytical models.

DAX operates on columns, tables and relationships within a semantic model. It combines concepts from Excel formulas, relational databases, and functional programming to provide powerful analytical capabilities.

It allows users to:

- Create calculated columns
- Create measures
- Create calculated tables
- Perform aggregation.
- Perform filtering
- Perform ranking
- Perform statistical analysis
- Perform time intelligence calculations
- Build KPIs

Components of DAX : DAX consists of three major calculation objects:

- **Calculated Columns:** Calculated columns compute values row by row and become permanent columns within the data model. Examples: Profit
- **Measures:** Measures perform calculations dynamically based on filters applied in reports. Examples: Total Sales.
- **Calculated Tables:** Calculated tables generate entirely new tables using DAX expressions. Example: Calendar Table

Characteristics of DAX: DAX is:

- Column-based
- Relationship-aware
- Context-sensitive
- Case-insensitive
- Functional
- Optimized for analytical calculations
- Integrated with VertiPac Engine

1.1 Create a measure to calculate the total sales amount from the sales table.

Answer:

- Open the Modeling tab
- Select New Measure
- Enter the DAX formula:
 - Total Sales = SUM(Sales[SalesAmount])
- Press Enter
- Add the measure to a Card visual

1.2 Create a measure that calculates the total quantity sold.

Answer:

- Create a New Measure
- Enter the DAX
 - $\text{Total Quantity} = \text{SUM}(\text{Sales}[\text{Quantity}])$
- Press Enter
- Create a Card visual and drag: Total Quantity to Card

1.3 Create a calculated column that computes Profit for each sales transaction.

Answer:

- Select the Sales table
- Click New Column
- Enter the DAX expression
 - $\text{Profit} = \text{Sales}[\text{SalesAmount}] - \text{Sales}[\text{Cost}]$
- Press Enter

1.4 Create a measure to calculate the average sales amount.

Answer:

- Open the Modeling tab
- Select New Measure
- Enter the DAX Formula
 - $\text{AverageSales} = \text{SUM}(\text{Sales}[\text{SalesAmount}])$
- Press Enter
- Add the measure to a Card visual

1.5 Create a measure that counts the total number of sales transaction.

Answer:

- In the Data/Fields pane on the right:
 - Find Sales
 - Click the Sales table
- Create a new measure
- Enter the DAX formula
 - $\text{TotalTransactions} = \text{COUNTROWS}(\text{Sales})$
- Create a Card visual
 - Go to Report view
 - Select Card
 - Drag Total Transactions into the Card

1.6 Create a measure that calculate the minimum sales amount recorded in the sales table.

Answer:

- Open the Modeling tab
- Select New Measure
- Enter the DAX Formula
 - `MinimumSales=MIN(Sales[SalesAmount])`
- Press Enter
- Add the measure to a Card visual

1.7 Create a measure that calculate the maximum sales amount.

Answer:

- Open the Modeling tab
- Select New Measure
- Enter the DAX Formula
 - `MaximumSales=MAX(Sales[SalesAmount])`
- Press Enter
- Add the measure to a Card visual

1.8 Create a measure to calculate the total profit.

Answer:

- Open the Modeling tab
- Select New Measure
- Enter the DAX Formula
 - `TotalProfit=SUM(Sales[Profit])`
- Press Enter
- Add the measure to a Card visual

1.9 Create a measure to calculate the average profit.

Answer:

- Open the Modeling tab
- Select New Measure
- Enter the DAX Formula
 - `AverageProfit=AVERAGE(Sales[Profit])`
- Press Enter
- Add the measure to a Card visual

1.10 Create a measure that counts the number of unique customers.

Answer:

- Open the Modeling tab
- Select New Measure
- Enter the DAX Formula
 - `TotalCustomers=DISTINCTCOUNT(Sales[CustomerID])`
- Press Enter
- Add the measure to a Card visual